

# Chapter 17: Complex Numbers, Powers & De Moivre's Formula

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**Course:** Applied Ordinary Differential Equations (ENGR 213) | **Textbook:** Official Course Textbook (7th Ed., Sections 17.1 & 17.2)

## 1. Executive Summary & First-Principles Philosophy

Complex numbers are the mathematical bridge that allows higher-order differential equations and vibrations to be solved elegantly. Without the imaginary unit  $i = \sqrt{-1}$ , oscillatory solutions like  $\sin(\omega t)$  and  $\cos(\omega t)$  would have to be handled through cumbersome trigonometric identities. Euler's formula  $e^{i\theta} = \cos \theta + i \sin \theta$  unifies algebraic exponentials with oscillatory trigonometry.

## 2. Core Mechanics & Mathematical Engine

### A. Rectangular and Polar Representations

- Rectangular Form:**  $z = x + iy$
- Modulus:**  $|z| = r = \sqrt{x^2 + y^2}$
- Argument:**  $\theta = \arg(z) = \text{atan2}(y, x)$  (Principal argument:  $-\pi < \text{Arg}(z) \leq \pi$ )
- Polar / Exponential Form:**

$$z = r(\cos \theta + i \sin \theta) = re^{i\theta}$$

### B. Powers & De Moivre's Formula

For any integer  $n$ :

$$z^n = (re^{i\theta})^n = r^n e^{in\theta} = r^n (\cos(n\theta) + i \sin(n\theta))$$

### C. Roots of Complex Numbers ( $n$ -th Roots)

To solve  $w^n = z_0 = r_0 e^{i\theta_0}$ , there exist exactly  $n$  distinct roots distributed symmetrically on a circle of radius  $r_0^{1/n}$  with angular spacing  $\frac{2\pi}{n}$ :

$$w_k = r_0^{1/n} \exp\left(i \frac{\theta_0 + 2k\pi}{n}\right), \quad k = 0, 1, 2, \dots, n-1$$

### 3. Curriculum-Grounded Visual Reference

**Addition:**  $z_1 + z_2 = (x_1 + iy_1) + (x_2 + iy_2) = (x_1 + x_2) + i(y_1 + y_2)$

**Subtraction:**  $z_1 - z_2 = (x_1 + iy_1) - (x_2 + iy_2) = (x_1 - x_2) + i(y_1 - y_2)$

**Multiplication:**  $z_1 \cdot z_2 = (x_1 + iy_1)(x_2 + iy_2)$   
 $= x_1x_2 - y_1y_2 + i(y_1x_2 + x_1y_2)$

**Division:**  $\frac{z_1}{z_2} = \frac{x_1 + iy_1}{x_2 + iy_2}$   
 $= \frac{x_1x_2 + y_1y_2}{x_2^2 + y_2^2} + i \frac{y_1x_2 - x_1y_2}{x_2^2 + y_2^2}$

Figure 17.1: The Argand complex plane: Cartesian coordinates  $(x, y)$  vs Polar  $(r, \theta)$  — from Textbook 7th Ed. Chapter 17 (Fig. 17.1.1).

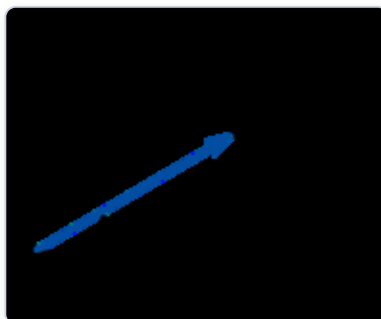


Figure 17.2: Roots of a complex number evenly spaced on a circle in the complex plane — from Textbook 7th Ed. Chapter 17 (Fig. 17.2.1).

### 4. Fully Worked Exam Archetype: Finding All Cube Roots

**Problem:** Find all cube roots of  $z = -8i$  in exact rectangular form.

#### Step-by-Step Solution:

- **Step 1: Convert  $z$  to polar form:** Modulus  $r = |-8i| = 8$ . Argument:  $-8i$  lies on the negative imaginary axis  $\implies \theta = -\frac{\pi}{2}$  (or  $\frac{3\pi}{2}$ ).

$$z = 8e^{-i\pi/2}$$

- **Step 2: Apply the  $n$ -th Root Formula for  $n = 3$ :**

$$w_k = 8^{1/3} \exp\left(i \frac{-\pi/2 + 2k\pi}{3}\right) = 2 \exp\left(i \left[-\frac{\pi}{6} + \frac{2k\pi}{3}\right]\right), \quad k = 0, 1, 2$$

- **Step 3: Evaluate each root:**

- **For**  $k = 0$ :

$$\theta_0 = -\frac{\pi}{6} \implies w_0 = 2 \left( \cos \left( -\frac{\pi}{6} \right) + i \sin \left( -\frac{\pi}{6} \right) \right) = 2 \left( \frac{\sqrt{3}}{2} - \frac{1}{2}i \right) = \sqrt{3} - i$$

- **For**  $k = 1$ :

$$\theta_1 = -\frac{\pi}{6} + \frac{2\pi}{3} = \frac{\pi}{2} \implies w_1 = 2 \left( \cos \left( \frac{\pi}{2} \right) + i \sin \left( \frac{\pi}{2} \right) \right) = 2(0 + i) = 2i$$

- **For**  $k = 2$ :

$$\theta_2 = -\frac{\pi}{6} + \frac{4\pi}{3} = \frac{7\pi}{6} \implies w_2 = 2 \left( \cos \left( \frac{7\pi}{6} \right) + i \sin \left( \frac{7\pi}{6} \right) \right) = 2 \left( -\frac{\sqrt{3}}{2} - \frac{1}{2}i \right) = -\sqrt{3} - i$$

## 5. Rapid Exam Traps & Red Flags

-  **Trap 1: Quadrant Checks in  $\arctan(y/x)$ :** Never blindly calculate  $\arctan(y/x)$ ! If  $z = -1 - i$ ,  $y/x = 1$ , but  $\arctan(1) = \pi/4$  (Quadrant I), whereas  $z$  is in Quadrant III. You must add  $\pi$  (or subtract  $\pi$ ) to get  $-\frac{3\pi}{4}$ .
-  **Trap 2: Forgetting Root Spacing:** The roots are separated by exactly  $\frac{2\pi}{n}$  radians. If your roots are not symmetrically distributed on a circle, your angular arithmetic has an error.